Agent Instructions: Selecting 20 Comparables per Company

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1. Objective and theory

For each of the 20 sample companies ("targets"), select exactly 20 listed comparable companies, compute the multiples below for each comparable, aggregate them, and apply the result to the target. The comparables must be chosen on **fundamentals** (growth, risk, profitability, reinvestment), using industry as the starting point but not as the only criterion.

Multiples to compute (course framework, Lecture 5):

- Earnings multiples: P/E (plus PEG and Relative P/E), EV/EBIT, EV/EBITDA, EV/Cash flow
- Book value multiples: P/BV, EV/Book value of assets, EV/Replacement cost (Tobin's Q, only if the data exists)
- Revenue multiples: P/S, EV/Sales

What makes a firm comparable. The course defines relative valuation as valuing an asset from the market prices of "similar" or "comparable" assets, and says that when assets are not fully comparable, "control" variables are needed (Lecture 1, slide 29). It lists three ways to build the group: identical firms, sector, and valuation-based comparables (Lecture 1, slide 30). The slides do not define "similar" further. The course's main textbook (Damodaran, Lecture 1 slide 36) defines a comparable firm as one with similar cash flows, growth potential and risk, and treats industry only as a proxy for these. This procedure combines sector and valuation-based selection.

Why fundamentals decide. Every multiple is a compressed DCF. Dividing a DCF model by earnings, book value or sales shows which fundamentals drive each multiple (Lecture 5, slides 11, 21–22, 28 and 37). A comparable must match the target on those drivers, otherwise the peer average is applied to a company that deserves a different multiple.

Multiple	Fundamentals that drive it	Source
P/E (and PEG, Relative P/E)	Expected growth, payout ratio, risk (cost of equity)	Lecture 5, slide 11 (P/E derived from the dividend discount model). PEG and Relative P/E: Damodaran
EV/EBIT, EV/EBITDA, EV/Cash flow	Expected growth and risk (cost of capital); also reinvestment rate and tax rate	Lecture 5, slides 21–22 (Value/FCFF). Reinvestment and tax rate: Damodaran
P/BV	ROE, growth, payout, risk	Lecture 5, slides 28 and 33
EV/Book value of assets, Tobin's Q	Return on capital, growth, risk	Damodaran (not in the slide text)
P/S	Net margin, growth, payout, risk	Lecture 5, slides 37 and 39
EV/Sales	Operating margin, growth, reinvestment, risk	Damodaran (not in the slide text)

The common drivers are growth, risk, profitability and reinvestment. These four, plus size and business model, are the selection criteria below.

2. Settings and definitions

Use one set of definitions for every company, target and comparable. A multiple is only comparable if numerator and denominator are measured the same way everywhere.

```
valuation_date: <one date for all prices, e.g. last close before the run>
earnings_basis: trailing FY2025       # also collect forward NTM where
available
comparables_per_target: 20
min_valid_values_per_multiple: 10
min_market_cap_eur: 1_000_000_000
size_band: [0.1, 10]                  # comparable market cap / target market
cap
reporting_currency: EUR               # convert everything at the
valuation_date FX rate
```

Item	Definition
Market value of equity	Share price × shares outstanding, at `valuation_date`
Enterprise value (EV)	Market value of equity + total debt (incl. IFRS 16 leases) + minority interests − cash and equivalents − equity-accounted associates. The course defines Value = market value of equity + market value of debt − cash (Lecture 5, slide 20); leases, minorities and associates are standard market practice added for consistency
EPS	Net income attributable to parent ÷ shares; same basis (trailing or forward) for all
EBIT, EBITDA	As reported; if a one-off item exceeds 10% of EBIT, also record the figure excluding it and flag it (our rule, not from the slides)
Cash flow (for EV/CF)	Operating cash flow, or FCFF if available for all comparables
Book value of equity	Equity attributable to parent; P/BV is not meaningful if negative
Book value of assets	Total assets (or invested capital, if used for all)
Expected growth	Analyst consensus EPS growth, next 3–5 years; fallback: fundamental growth = ROE × (1 − payout), i.e. retention ratio × ROE (Lecture 3, slide 58)
PEG	P/E ÷ expected growth in percent (P/E 20, growth 10% → PEG 2.0)
Relative P/E	Company P/E ÷ P/E of its home market index

Why these rules. EV must include everything a buyer of the operations would pay for (debt, leases, minorities) and exclude what is not operating (cash, associates); otherwise EV multiples are biased for highly leveraged utilities. A common valuation date and currency remove timing and FX noise. Fiscal years that do not end in December (for example SSE in March, Siemens Energy in September) must be calendarised to the nearest year end, or flagged.

Consistency tests from the course (Lecture 5, slide 20): if EV is net of cash, interest income on that cash must not be in the denominator; the debt included in EV must match the interest added back to reach EBIT. For P/BV, use the book value of common equity when the market value refers to common stock, and include all share classes (Lecture 5, slide 26).

3. The selection funnel

Run these steps separately for each target. Start wide (40–60 candidates), cut with hard filters, rank by fundamental similarity, then keep the best 20 under composition rules.

Source note. The slides set the principles (comparable assets, sector or valuation-based groups, control variables: Lecture 1, slides 29–30) but give no step-by-step selection procedure. The tiers, filters, size band and composition rules below are our method for applying those principles; present them as such in the report.

1. **Profile the target.** Record its main activity (the segment with at least 50% of revenue or EBITDA), sub-industry, and the metrics in section 4 (growth, beta, net debt/EBITDA, ROE, EBITDA margin, payout, capex/sales, market cap).
2. **Build the candidate pool in three tiers.**
   - **Tier 1, same business:** firms whose main activity is the same as the target's.
   - **Tier 2, adjacent business with similar economics:** firms in a neighbouring industry with the same cash-flow profile (regulated or contracted vs cyclical; similar capital intensity).
   - **Tier 3, any other non-financial firm:** used only if Tiers 1–2 cannot supply 20 comparables after filtering.
   - Geography: Europe first, then other developed markets (North America, Japan, Australia).
   - Start from the target's "Peer list" in the valuation workbook, then extend with your data source's screener.
3. **Apply the hard filters.** Remove a candidate if any of these holds:
   - it is in an emerging market (MSCI classification);
   - it is a bank, insurer or holding company;
   - market cap is below EUR 1bn or outside 0.1×–10× the target's;
   - it is the target itself, its parent, its subsidiary, or holds or is held by the target with a stake above 20%;

- it had a merger, spin-off or restructuring in FY2024–25 that makes its financials non-comparable;
- it has less than 2 years of trading history (no reliable beta);
- more than 2 of the section 4 metrics are missing.

4. **Score every remaining candidate** with the similarity score in section 4.
5. **Pick the 20 with the lowest score, subject to these composition rules:**
   - at least 12 from Tiers 1–2;
   - at most 4 from Tier 3;
   - at least 10 listed in Europe. If a rule is violated, replace the worst-ranked comparable that breaks it with the best-ranked candidate that satisfies it.
6. **If fewer than 20 qualify, relax in this order** and log each step: (a) add non-European developed markets; (b) widen the size band to 0.05×–20×; (c) allow up to 8 Tier 3 firms. Never relax the other filters.
7. **Review by reading.** Read each selected firm's business description. Replace any firm whose business is clearly unrelated despite similar numbers (for example a software company matched to a utility on growth and margin) with the next-ranked candidate, and log the reason.

Why this design.

- Tiers keep industry as the starting point without making it a requirement: same-industry firms share accounting, cycle and regulation, but the course defines comparability by cash flows, growth and risk, not by sector label.
- Emerging markets are excluded because country risk alone shifts multiples: in the course's Telebras regression, the emerging-market dummy lowers the predicted P/E by about 13.9 points (Lecture 5, slide 17). Mixing them in would bias the peer average downward.
- The size band avoids comparing a EUR 100bn utility with a EUR 1bn developer; size proxies liquidity, diversification and access to capital.
- Parent and subsidiary pairs (for example EDP and EDP Renováveis) share the same cash flows, so including one as a comparable for the other double-counts the same information.
- The composition rules stop the score from filling the list with firms that match on numbers but come from unrelated businesses.

Tier examples per cluster (names are starting points; verify listing status and filters before use):

Target type	Tier 1 (same business)	Tier 2 (adjacent)
Renewable power producers (EDPR, Ørsted)	Listed renewable developers and IPPs (e.g. Acciona Energía, ERG, Solaria, Brookfield Renewable)	Integrated utilities with large renewable fleets (e.g. RWE, Iberdrola, Enel)
Integrated utilities (Iberdrola, Enel, SSE, Engie, Verbund)	European integrated and regulated utilities (e.g. E.ON, EDP, Endesa, Naturgy, Fortum, National Grid)	Grid operators (e.g. Terna, Redeia, Elia) and developed-market integrated utilities outside Europe
Wind and power equipment (Vestas, Siemens Energy)	Power generation equipment (e.g. Nordex, GE Vernova, Mitsubishi Heavy Industries)	Electrical equipment (e.g. ABB, Schneider Electric, Hitachi, Eaton)
Electrical equipment and cables (Schneider, Prysmian, NKT)	Same product line (cables: Nexans, Sumitomo Electric, Furukawa Electric; electrical equipment: ABB, Legrand, Eaton)	The other product line, then diversified capital goods
Oil and gas majors (TotalEnergies, Shell, BP, Equinor)	Integrated majors (e.g. Eni, Repsol, OMV, ExxonMobil, Chevron)	Large E&P and LNG companies
Renewable fuels (Neste)	Renewable fuels and refiners (e.g. Darling Ingredients, Valero, Galp, Orlen)	Integrated oil companies with refining exposure
Industrials (Alfa Laval, Air Liquide, ArcelorMittal)	Own sub-industry (gases: Linde, Air Products, Nippon Sanso; machinery: GEA, Atlas Copco, Wärtsilä, Xylem; steel: Nucor, voestalpine, SSAB, thyssenkrupp)	Adjacent industrial sub-industries with similar cyclicality

Note: some names in the workbook's peer lists (for example Goldwind, Suzlon, POSCO, LS Corp) are in emerging markets and are removed by step 3.

4. Similarity score

The score measures how far a candidate's fundamentals are from the target's. Lower is more similar. Industry is not in the score: it is handled by the tiers and composition rules in section 3.

Dimension	Metric	Weight	Fallback if missing	Why it matters
Growth	Expected EPS growth, next 3–5 years (consensus)	30%	ROE × (1 − payout)	Appears in the determinants of every multiple; significant in the course's P/E (Lecture 5, slide 17) and P/S (slide 39) regressions, but in the P/BV regression only for the US (slide 33)
Risk	5-year levered beta	15%	Bottom-up beta: industry unlevered beta (Damodaran, Europe) relevered at the firm's D/E (Lecture 2, slides 64–72)	Sets the cost of equity, which drives P/E, P/BV and P/S
Risk	Net debt / EBITDA	10%	Net debt / equity	Financial risk; separates highly levered developers from conservative peers
Profitability	ROE	10%	Net income / average equity from the last two years	Drives P/BV; also proxies the quality of growth

Dimension	Metric	Weight	Fallback if missing	Why it matters
Profitability	EBITDA margin	10%	EBIT margin	Drives EV/Sales and P/S; distinguishes asset-heavy from asset-light models
Reinvestment	Payout ratio	7.5%	DPS / EPS	Links growth to cash returned; drives P/E and P/BV
Reinvestment	Capex / sales	7.5%	(Capex − D&A) / sales	Reinvestment needs drive EV multiples
Size	ln(market cap in EUR)	10%	None (required)	Proxies liquidity, diversification and access to capital

How to compute it (standard statistics; the slides do not prescribe a scoring method).

1. For each metric, winsorize the candidate pool at the 5th and 95th percentiles, so a single extreme value cannot dominate.
2. Standardize: z = (value − pool mean) ÷ pool standard deviation. Compute the target's z with the same mean and standard deviation.
3. Score each candidate:

```
D = \sqrt{\sum_k w_k \, (z_{k,\text{candidate}} - z_{k,\text{target}})^2}
```

4. If one or two metrics are missing for a candidate, drop them from its score and rescale the remaining weights to sum to 100%. Flag the candidate.

Why these weights. Growth gets the largest weight because it appears in the determinants of every multiple and is the variable that differs most across energy-transition firms (a regulated utility at 3–5% versus a wind OEM in recovery). Risk comes second because it sets the discount rate; it is split between market risk (beta) and financial risk (leverage) because betas in this sample are often unreliable (negative or near zero for oil majors, very low for Verbund). Profitability and reinvestment together capture how much growth costs, which is the "quality of growth" effect from the P/E fundamentals. Size gets a small weight because it is partly already reflected in beta and leverage.

Theory versus the course's evidence on risk. Theory says higher risk lowers every multiple. Yet the course's own P/BV regression shows a positive beta coefficient for the US (0.358) and Europe (0.820), negative only for emerging markets (Lecture 5, slide 33), and Lecture 2 (slide 63) warns that regression betas have high standard errors. This is why risk is measured with both beta and leverage, and why beta carries only 15% of the weight.

The weights are our reasoned default; they are not taken from the slides. Record them in the output; if the user changes them, re-run the full selection.

5. From comparables to multiples and value

Use the **median** of the 20 comparables as the peer multiple, the interquartile range for the value range, and adjust for the target's own fundamentals before applying it.

Step A – Compute and clean. Compute all multiples from section 1 for each comparable. Mark a value "not meaningful" (NM) and leave it out of the statistics if its denominator is zero or negative (losses, negative EBITDA, negative book value). Keep high but positive values; the median handles them. A multiple is used for a target only if at least 10 of its 20 comparables have a valid value.

Step B – Aggregate. For each target and multiple, report: number of valid values, median, mean, 25th percentile and 75th percentile.

- Use the **median** as the peer multiple, not the simple mean. Multiples cannot go below zero but have a long right tail, so a few very high values pull the mean up. Lecture 1 (slide 30) speaks of the "peer group average", assuming the average is representative; the median is the robust form of that average, and Lecture 5 (slide 10) itself compares markets by median P/E (Europe 15.09).
- Report the mean for reference only. If mean and median differ by more than 25%, say so: the peer group is skewed.

Step C – Adjust for differences ("peer group average adjusted for differences", Lecture 1, slide 30). Even good comparables differ from the target on the main driver. Scale the multiple by that driver (these scaled ratios are Damodaran's textbook approach; the slides apply the same idea through regressions):

Multiple	Adjusted multiple for the target
P/E	Median peer PEG × target expected growth (in %)

Multiple	Adjusted multiple for the target
P/BV	Median peer (P/BV ÷ ROE) × target ROE
P/S	Median peer (P/S ÷ net margin) × target net margin
EV/Sales	Median peer (EV/Sales ÷ EBITDA margin) × target EBITDA margin
EV/EBITDA, EV/EBIT, EV/CF	No scaling; report the unadjusted median and the target's growth versus the peer median growth

Report both the unadjusted and the adjusted value. Optionally, regress the primary multiple on its main driver across the 20 comparables, as in the course examples (P/BV on ROE for European banks, Lecture 5 slide 29; P/S on net margin for European retailers, slide 38) (one variable only; with 20 observations, more variables are unreliable).

Step D – Value the target.

- Equity multiples (P/E, P/BV, P/S): value per share = multiple × target's per-share metric.
- Enterprise multiples (EV/EBIT, EV/EBITDA, EV/CF, EV/Sales, EV/assets): equity value = multiple × target metric − debt (incl. leases) − minorities + cash + associates; then divide by shares.
- Convert to the target's share-price currency.
- Point estimate from the median; range from the 25th–75th percentile multiples. The coursework asks for a fair value range and a recommended price.

Step E – Choose the primary multiple. The course recommends one multiple as the best estimate, with the others as cross-checks (Lecture 5, slide 41). Default choice by business type (our judgment; the slides do not assign multiples to sectors):

Target type	Primary multiple	Cross-checks	Why
Renewable power producers	EV/EBITDA	P/BV	Heavy depreciation and asset-rotation gains make net income volatile (EDPR and Ørsted had losses in FY2024); EBITDA tracks contracted cash generation

Target type	Primary multiple	Cross-checks	Why
Integrated utilities	EV/EBITDA	P/E, P/BV	Stable earnings; regulated asset base is close to book value
Wind and power equipment	EV/EBIT	EV/Sales, P/E	Margins are still recovering, so revenue multiples help when earnings are thin
Electrical equipment and cables	P/E	EV/EBIT	Profitable, asset-light, earnings close to cash earnings
Oil and gas majors	EV/EBITDA	EV/Cash flow, P/E	Commodity cycle makes earnings swing; cash flow multiples are standard for the sector
Renewable fuels (Neste)	EV/EBITDA	EV/Sales	FY2024 loss and thin FY2025 earnings make P/E unusable
Industrial gases and machinery	P/E	EV/EBIT	Stable, profitable, low leverage
Steel (ArcelorMittal)	EV/EBITDA	P/BV	Deeply cyclical and asset-heavy

If the target's primary-multiple denominator is negative or near zero, switch to the first cross-check and log why.

6. Output, quality checks and stop rules

Deliver three files: one row per comparable, one row per target-multiple pair, and a short report.

1. **`comparables.csv`** — one row per target × comparable (400 rows):

`target_ticker, rank, comp_name, comp_ticker, comp_isin, country, tier, distance_score, growth, beta, net_debt_ebitda, roe, ebitda_margin, payout, capex_sales, market_cap_eur, pe, peg, relative_pe, ev_ebit, ev_ebitda, ev_cf, pbv, ev_assets, tobins_q, ps, ev_sales, flags, inclusion_reason, source_urls`

2. **`multiples_summary.csv`** — one row per target × multiple:

`target_ticker, multiple, n_valid, median, mean, p25, p75, adjusted_multiple, value_per_share_low, value_per_share_mid, value_per_share_high, currency, is_primary`

3. **`comparables_report.md`** — for each target:

- the primary multiple and why;
- the value range and point estimate;
- the target's fundamentals against the peer medians (growth, beta, ROE, margin);
- every relaxation, replacement and flag.

Quality checks before delivery:

- [ ] Each target has exactly 20 comparables (or at least 15 with a logged reason)
- [ ] No target appears in its own list; no parent, subsidiary or cross-holding above 20%
- [ ] No emerging-market, bank, insurer or holding company
- [ ] Composition rules met: at least 12 from Tiers 1–2, at most 4 from Tier 3, at least 10 European
- [ ] Every ISIN verified on an exchange, MSCI or company IR page
- [ ] All prices at the same `valuation_date`; all values converted at the same FX date
- [ ] Every multiple used has at least 10 valid values
- [ ] EV hand-checked for 3 random comparables
- [ ] Every comparable has a one-line inclusion reason, and every figure has a linked source (section 7)

Never:

- invent or estimate a missing figure (use the stated fallbacks or leave it blank);
- relax the earnings, data-quality, emerging-market or financials filters;
- mix trailing and forward earnings in the same statistic.

Stop and report to the user if:

- a target still has fewer than 15 comparables after all relaxations;
- expected growth is missing for more than 30% of a candidate pool (the fallback would then dominate the score);
- the data source cannot provide prices at a single valuation date.

7. Source linking (mandatory)

Every figure and every judgment you use in this relative valuation must be traceable through a link. A figure without a linked source counts as missing: do not use it.

- **Link each figure.** For every value you collect (price, shares, EPS, net income, EBIT, EBITDA, debt, cash, book value, sales, beta, expected growth, payout, ISIN, MSCI classification, FX rate), record the full URL of the exact page or document it came from and the access date, in the `source_urls` column. Format: `field: URL (accessed YYYY-MM-DD)`, entries separated by `|`.
- **Link the exact page, not a home page.** For example the annual report PDF plus page number, or the exchange's quote page for that share.
- **Terminals without a public URL** (Bloomberg, Eikon): write the terminal, the ticker, the field or screen used, and the date.
- **Conflicting sources:** if two sources differ by more than 2%, link both and state which one you used and why.
- **Report:** end each target's section in `comparables_report.md` with a "Sources" list of every link used for that target. Cite course material as (Lecture X, slide Y).
- **Never** cite a source you did not actually open, and never write a URL from memory.